IDAT7215: Computer Programming for Product Development and Applications
(Mid Term Exam 23rd October 2025)

Student Name: Student ID:

Part I: (80 marks)

Question 1 (60 Marks)

Choose only one of the most appropriate answers for the following questions:

1. Which programming language was originally developed for UNIX?

(a) Python (b) C (c) Java (d) JavaScript

2. Which of the following is a basic data type in Python?

(a) Array (b) List (c) int (d) Hash

3. Which command is used to run a cell in Jupyter Notebook?

(a) Shift + Enter (b) Ctrl + Enter (c) Alt + Enter (d) Enter

4. Which statement is used for error handling in Python?

(a) try-except (b) if-else (c) for loop d) switch-case

5. What is the output of 15 % 2 in Python?

(a) 1 (b) 7 (c) 0 (d) 15

6. Which of the following is a mutable data type in Python?

(a) Tuple (b) String (c) Set (d) All of the above

7. In Python, what is the result of 4**2?

(a) 8 (b) 16 (c) 12 (d) 2

8. How do you denote a comment in Python?

(a) // (b) # (c) /* ... */ (d) --

9. Which statement is used to exit a loop prematurely in Python?

A) continue (b) exit (c) break (d) stop

10. What does the iloc attribute in Pandas do?

(a) Access data by label
(b) Access data by position
(c) Filter data
(d) None of the above

11. What is the output of list(range(5))?

(a) [1, 2, 3, 4, 5] (b) [0, 1, 2, 3, 4] (c) [0, 1, 2, 3, 4, 5] (d) None

12. What is the output of np.linspace(0, 1, 5)?

(a) [1, 2, 3, 4, 5] (b) [0, 1, 2, 3, 4] (c) [0, 0.25, 0.5, 0.75, 1] (d) [0, 0.2, 0.4, 0.6, 0.8]

13. Which function is used to find the number of elements in an array?

(a) length() (b) count() (c) size() (d) total()

14. In Matplotlib, which function is used to display a plot?

(a) show() (b) display() (c) plot() (d) draw()

15. Which method is used to get the first few rows of a DataFrame?

(a) df.head() (b) df.first() (c) df.top() (d) df.peek()

16. In which scenario would you prefer using a list over a tuple in Python?

(a) When you need a collection that can be modified after creation.
(b) When you need to store a collection of unique elements.
(c) When you want to create a fixed-size collection of elements.
(d) When you want to return multiple values from a function.

17. A control system is stochastic if:

(a) The response to input is unpredictable. (b) The response to input is predictable and repeatable.
(c) The output varies linearly with the input (d) None of these.

Answer: (a) A control system is stochastic if the response to the input is not predictable (see class lecture week3_1_control.pdf for details)

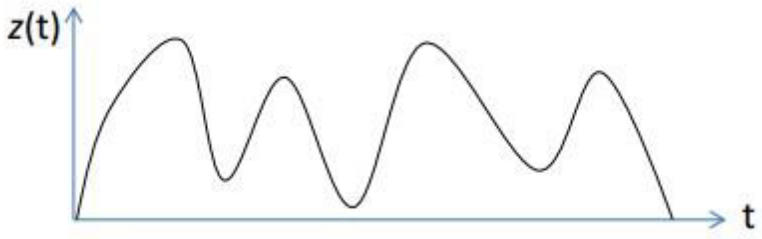


Figure 1: Output of Stochastic Control System

18. What does the transfer function represent in a control system?

(a) Relationship between input and output. (b) The physical layout of the system.
(c) The stability of the system. (d) The cost of the system

19. What does the term "BIBO" stand for in control systems?

(a) Bounded Input Bounded Output. (b) Basic Input Basic Output.
(c) Balanced Input Balanced Output. (d) Binary Input Binary Output.

20. Which of the following best describes a deterministic control system?

(a) The output is unpredictable.
(b) The output is predictable and repeatable.
(c) It requires random inputs.
(d) It is nonlinear.

21. What is the main advantage of depth-first search (DFS)?

(a) It is guaranteed to find the shortest path.
(b) It has low memory requirements.
(c) It always finds a solution.
(d) It expands all nodes at the same depth first.

22. Which algorithm combines the advantages of DFS and BFS?

(a) Bidirectional search. (b) Iterative deepening search. (c) Greedy search. (d) A* search.

23. Which of the following is NOT a reason for developing Expert Systems?

(a) Human expertise is in short supply.
(b) They are cheaper than hiring experts.
(c) They can replicate common sense.
(d) They can be used to preserve human knowledge.

24. What does the knowledge base of an Expert System contain?

(a) Factual and heuristic knowledge
(b) Only user input
(c) Only heuristic knowledge
(d) Programming code

25. Which of the following describes forward chaining?

(a) Starting with a known conclusion
(b) Moving from data input to a conclusion
(c) Asking questions based on past outcomes
(d) Simple data retrieval

26. Why was PyKE chosen for Expert Systems?

(a) It operates exclusively in C
(b) It is compatible with Python 3
(c) It is outdated
(d) It is harder to use than other systems

27. The key difference between supervised and unsupervised machine learning is

(a) Supervised learning requires labeled data, while unsupervised learning does not.
(d) Supervised learning does not require labeled data, while unsupervised learning does require
(b) Supervised learning predicts labels, while unsupervised learning discovers patterns.
(d) Supervised learning is always more accurate than unsupervised learning.

28. Which of the following is not a supervised machine learning algorithm?

(a) K-means (b) Naïve Bayes (c) Support Vector Machine (d) All

29. What is the output of Logistic Regression?

(a) Continuous values (b) Categorical values (c) Time series data (d) Image data

30. What does the F1 score measure in a classification model?

(a) The model's accuracy.
(b) The balance between precision and recall.
(b) The total number of predictions.
(d) The speed of the model.

Question 2 (20 Marks)

True/False questions.

1. Linkers do not handle library files; they only work with object files generated from source code.
   (a) True (b) False
   Answer: Linkers can also handle library files.

2. The interpreter executes faster compared to the compiler.
   (a) True (b) False
   Answer: The compiler translates the entire source code into machine code before execution. In contrast, the interpreter translates code line by line during execution, which makes it slower compared to the compiler.

3. A Python set can contain duplicate elements.
   (a) True (b) False
   Answer: The Sets are collections of unique elements.

4. Logical errors are found when the program runs to completion without error but produces the wrong output.
   (a) True (b) False

5. You can concatenate multiple strings at once by using the + operator in a single expression.
   (a) True (b) False

6. A loop can be nested inside another loop in programming.
   (a) True (b) False

7. In a control system, a controller is responsible for comparing the desired output to the actual output.
   (a) True (b) False

8. A control system is deterministic if the response to input is predictable and repeatable.
   (a) True (b) False

9. If all the poles of the system lie in the left half plane, the system is said to be unstable.
   (a) True (b) False
   Answer: If all the poles of the system lie in the left half plane the system is said to be stable.

10. Breadth-first search (BFS) explores all nodes at the present depth before moving on to nodes at the next depth level.
    (a) True (b) False

11. The complexity of a search algorithm is not affected by the branching factor of the search tree.
    (a) True (b) False
    Answer: The Complexity of a search algorithm is dependent on the branching factor of the search tree.

12. Bidirectional search runs two simultaneous searches, one from the start state and one from the goal state.
    (a) True (b) False

13. Heuristic functions are used in AI to guarantee the optimality of a search algorithm
    (a) True (b) False
    Answer: Heuristic functions provide estimates but do not guarantee optimality

14. An expert system is a computer program that simulates the decision-making ability of a human expert.
    (a) True (b) False

15. Expert systems are only as good as the data and rules they are given.
    (a) True (b) False

16. The inference engine is responsible for reasoning and concluding the knowledge base.
    (a) True (b) False

17. Cross-validation is a technique used to prevent underfitting in machine learning models.
    (a) True (b) False
    Answer: It helps prevent overfitting.

18. Supervised learning algorithms can be used for both classification and regression tasks.
    (a) True (b) False

19. Support Vector Machine (SVM) and Random Forest algorithms are examples of unsupervised machine learning.
    (a) True (b) False
    Answer: They are an example of supervised learning.

20. Recommendation systems are a classic example of unsupervised machine learning.
    (a) True (b) False

Part II: (20 marks)

Question 3 (10 marks):

Answer the following questions about Python Programming.

3.1 What will be the output of the following Python code? **(2 marks)**

```
a = [1, 14, 3, 16, 18, 4, 19, 17, 7, 6, 11, 15, 8, 10, 2]
a.sort(reverse=True)
print(a[9:3:-3])
```

Ans: [7,11]

3.2 What will be the output of the following Python code? **(2 marks)**

```
import numpy as np
b= np.arange(16).reshape(4,4)
row = np.array([2,0])
col = np.array([3,3])
print(b[row,col])
```

Ans: [11 3]

3.3 What will be the output of the following Python code? **(2 marks)**

```
a = [1, 6, 3, 9, 7, 8]
a.sort(reverse=True)
print(a[5:2:-2])
```

Ans: [1, 6]

3.4 What will be the output of the following Python code? **(2 marks)**

```
def my_print(person = "Joe"):
    message = "Hello, " + person
    print(message)

my_print("Peter")
```

Ans: Hello, Peter

3.5 What will be the output of the following Python code? **(2 marks)**

```
import pandas as pd
df = pd.DataFrame([[1000, "Jack", 21], [1500, "John", 29]], columns=["1", "2", "3"])
df
```

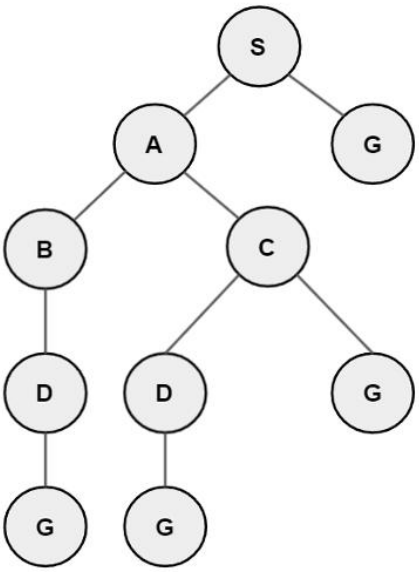
	1	2	3
0	1000	Jack	21
1	1500	John	29

```
print(df.iloc[1,1])
```

Ans: John

Question 4 (4 marks)

Look at the graph below and traverse nodes in the tree starting from node "S" using:
1) **breadth-first search**; 2) **depth-first search**.



ANSWER

1) breadth-first search: S,A,G, B,C,D,D,G,G,G
2) depth-first search: S,A,B,D,G, C,D,G,G,G

Alternative answers are also acceptable:
1) breadth-first search: S,A,G
2) depth-first search: S,A,B,D,G

Question-5 (6 marks)

Suppose you have the following rules in your knowledge base, and we are using forward chaining (data-driven), what "goal" would the system return if A, B, C, and E are true? Example of how the results are obtained (e.g., show the reasoning path).

Rule 1: IF X is true
AND B is true
AND E is true
THEN Y is true

Rule 2: IF Y is true
AND D is true
THEN Z is true

Rule 3: IF A is true
THEN X is true

Rule 4: IF Y is true
AND C is true
THEN W is true

Answer:
The system will return "W" as the goal. Since A is true then by following Rule 3 X is true. Now if X, B, and E are true then following Rule 1 Y is true. Now If Y and C are True then following Rule 4 W is true. Rule 2 is not applicable as we do not know the status of D. So, the reasoning path is.

Rule 3 $\Longrightarrow$ Rule 1 $\Longrightarrow$ Rule 4.

END OF EXAM-